

Continuous glucose monitoring use and glucose variability in very young children with type 1 diabetes (VibRate): A multinational prospective observational real-world cohort study

Abstract

Precise management of type 1 diabetes (T1D) for very young children is crucial yet challenging. Continuous glucose monitoring (CGM) has shown its efficacy except for very young children. This study aims to find whether applying CGM is associated with diminishing glucose excursions and increasing time spent in range (TIR, 70-180 mg/dl [3.9-10.0 mmol/L]). The open-label, prospective, observational, multinational, registry-based population cohort design was adopted to compare glycemic metrics over 12 months among 339 young children of 1-7 years old with T1D who used CGM and those used finger stick blood glucose monitoring alone (BGM). The primary endpoint was to compare glycemic variability difference between these two cohorts by coefficient of variation (CV). Other endpoints included standardized glycemia metrics. The CV among CGM and BGM users were 39.1% and 46.8%, respectively. The proportion of TIR was 65.2% and 46.8% in the CGM and BGM groups, respectively, with less pronounced excursions in the ambulatory glucose profile for the former. The median HbA1c was 7.0% and 7.5% for the CGM and BGM groups, respectively. 33.7% (59 users) of the CGM achieved the clinical target of TIR more than 70%, comparing to the 3.8% (2) for the BGM. One severe hypoglycemia event occurred in the CGM and four among the BGM cohort. Diabetic ketoacidosis occurred twice in one CGM user. In conclusion, CGM was found to be associated with reduced glycemic variability for very young children. Youth using CGM spent about 15% more TIR than non-users.

Key Points

1. The largest real-world analysis of glycemic metrics in young children with T1D using CGM to date.
2. CGM use was associated with improved glycemic control among very young children with T1D.
3. CGM use could have long-term clinical benefit for very young children with T1D.
4. Long-term studies are needed to assess CGM outcomes, especially with semi-automated insulin delivery.